

Performance Analysis of M-PAM with Matched Filtering

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Abstract

This report investigates the performance of a digital communication system using M-ary Pulse Amplitude Modulation (M-PAM). The system's Bit Error Rate (BER) and Symbol Error Rate (SER) are analyzed over an Additive White Gaussian Noise (AWGN) channel. We compare theoretical results with simulations performed using both MATLAB scripts and Simulink models. To observe the effect of pulse shaping, three different filters are evaluated: Rectangular, Raised Cosine, and Triangle.

1 Introduction

The problem of recovering signals corrupted by noise is a key problem in digital communications. The correlation receiver is known to be the best and optimal solution for recovering a known waveform in the presence of additive white Gaussian noise (AWGN). A correlation receiver works by comparing the received signal to a set of waveforms it expects to receive. This makes sense for digital communications, as each bit or sequence is encoded into a distinct waveform known to both the sender and the receiver.

A correlation receiver parses the signal into intervals synchronized with the bits, and integrates the product of the received signal and the templates. This produces correlation values, which act as detection statistics to decide the most likely transmitted bit. If we flip (time-reverse) the template and use it as a filter's impulse response, the filter operation essentially reverses it back and performs a correlation. This process is called matched filtering, and it yields the exact same optimal outcome as a correlation receiver. In this project, i explored the design principles of matched filters and evaluate the probability of error using MATLAB and Simulink.

2 System Model and Methodology

To fully understand the digital channel, i simulated a baseband M-PAM communication link. I considered both the standard 2-level PAM and higher-order M-level PAM (specifically 4-PAM and 8-PAM).

2.1 Simulink Model

The system was first modeled in Simulink. Random integers are generated, converted into bits, and mapped into M-PAM symbols. These symbols are passed through a transmit pulse-shaping filter, transmitted over an AWGN channel, and then processed by a receive matched filter. Finally, they are demodulated so that the BER and SER can be calculated.

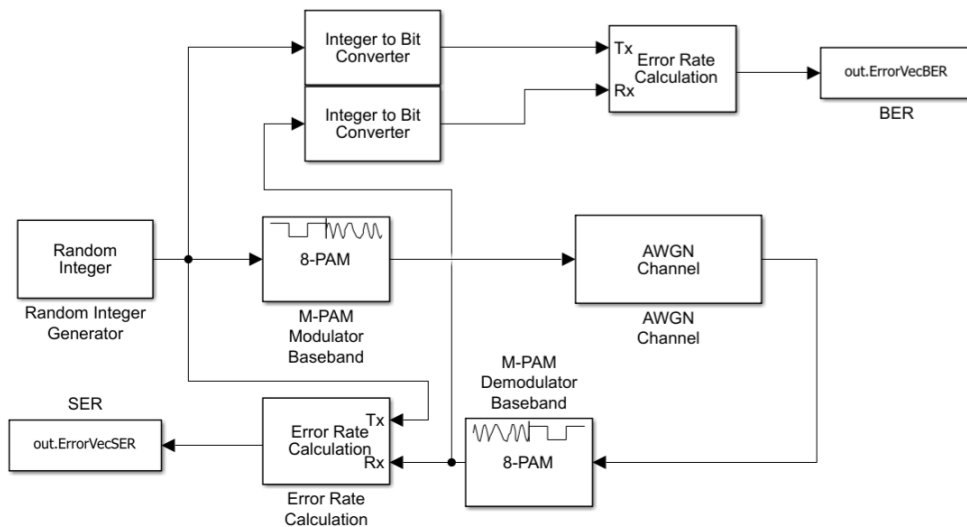


Figure 1: Block diagram of the M-PAM baseband communication system simulated in Simulink.

2.2 MATLAB Simulation

To verify the Simulink model, i repeated the simulations using standard MATLAB scripts. The simulation sweeps through Signal-to-Noise Ratio (SNR) values defined by E_b/N_0 from 0 dB to 20 dB. Three different pulse shapes were tested at 8 samples per symbol: a basic Rectangular pulse, a Square-Root Raised Cosine pulse (with a roll-off factor of 0.5), and a Triangle pulse.

3 Results and Discussion

When analyzing a digital system, a bit error occurs when noise or bandwidth limits cause an incorrect decision. The number of errors per bit during our simulation is our Bit Error Rate (BER). As expected, the probability of detection increases (and the error rate drops) as the SNR increases.

3.1 Rectangular Pulse Shape

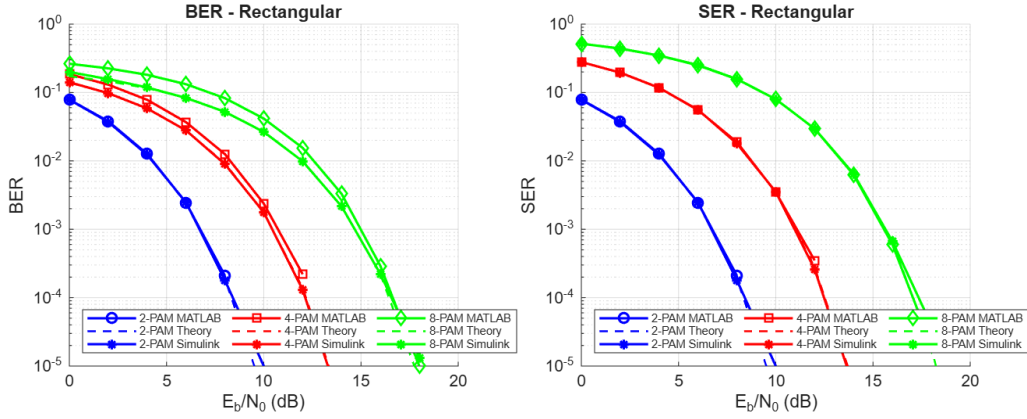


Figure 2: BER and SER performance for 2-PAM, 4-PAM, and 8-PAM using a Rectangular pulse.

Figure 2 shows that the MATLAB and Simulink simulations perfectly match the theoretical probability of error curves. While the rectangular pulse is very simple to implement, it usually requires a larger bandwidth in practical, real-world applications.

3.2 Raised Cosine Pulse Shape

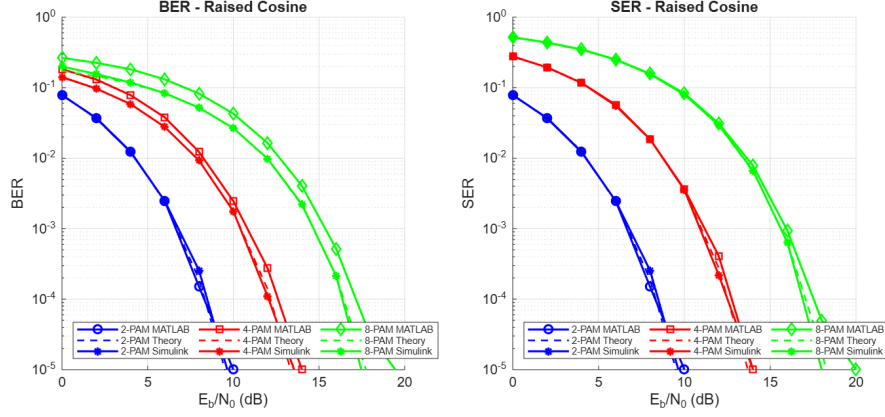


Figure 3: BER and SER performance for 2-PAM, 4-PAM, and 8-PAM using a Raised Cosine pulse.

Figure 3 highlights the system's performance using the Square-Root Raised Cosine filter. This pulse shape is widely used because it minimizes Inter-Symbol Interference (ISI). Once again, our simulated data for all M-levels strictly follows the theoretical formulas.

3.3 Triangle Pulse Shape

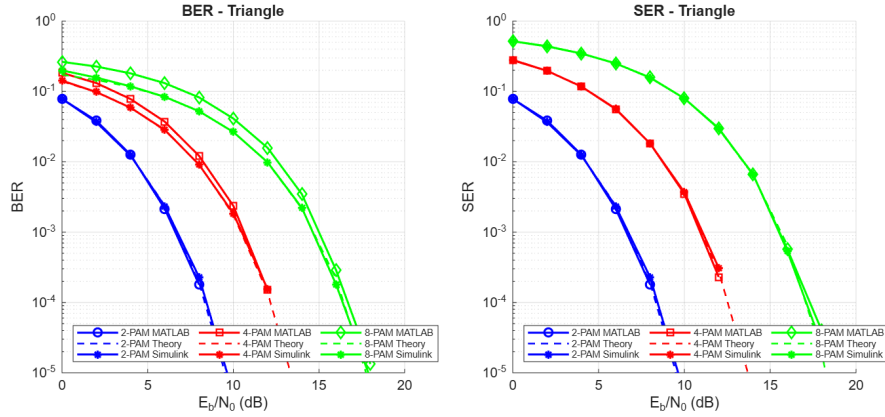


Figure 4: BER and SER performance for 2-PAM, 4-PAM, and 8-PAM using a Triangle pulse.

Figure 4 shows the results for the triangular pulse shape. By pushing custom filter coefficients into our Simulink model, we were able to match the convolution output of the MATLAB script. Across all figures, we clearly see that higher-order modulations (like 8-PAM) need a much higher SNR to achieve the same BER as 2-PAM because the symbols are packed closer together.

4 Conclusion

This project successfully confirmed the equivalence of correlation receivers and matched filtering in AWGN channels. By simulating 2-PAM, 4-PAM, and 8-PAM, we observed the fundamental trade-off in digital communications: higher data rates (larger M) require significantly more energy to maintain reliable detection. The tight alignment between our theoretical formulas, MATLAB code, and Simulink blocks validates the accuracy of our system design and error rate calculations.

References

- [1] CIE 237- Course Materials, 2026.